

SUMMER INTERNSHIP REPORT

GenAI Powered Data Analytics

Tata iQ • Forage Virtual Job Simulation

Submitted by	Ayush Shekhar
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Academic Credit	1 Credit

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Prepared for academic internship submission

University IILM University

University Roll No. 25SCS1003002962

DECLARATION

I, Ayush Shekhar, a student of B.Tech. Computer Science & Engineering (Data Science), Batch 2025-2029, declare that this internship report is prepared on the basis of my participation in the Tata iQ GenAI Powered Data Analytics job simulation hosted on Forage. The report summarizes the activities, concepts, tasks and learning outcomes documented during the simulation. The work is presented for academic internship/credit submission and does not claim employment with Tata or Forage.

Student Name: Ayush Shekhar

Signature: _____

Date: September 20, 2026

oI 25SCS1003002962

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Tata iQ and Forage teams for providing a structured virtual job simulation in GenAI-powered data analytics. The simulation offered an opportunity to work through a realistic financial-services scenario involving customer delinquency, exploratory data analysis, predictive modeling and responsible AI.

I also thank my college faculty and department for encouraging practical learning through internship and credit-based activities. The experience helped me connect classroom concepts such as data analysis, machine learning, exploratory analysis and model evaluation with a business-oriented problem.

Finally, I acknowledge the value of GenAI tools as learning and productivity assistants while recognizing that analytical outputs require validation, domain context, fairness checks and human review.

EXECUTIVE SUMMARY

This internship experience was completed through the GenAI Powered Data Analytics job simulation on Forage, in which the participant works in the simulated role of an AI transformation consultant at Tata iQ. The official program describes it as a self-paced 3–4 hour experience covering predictive modeling using GenAI, data exploration and quality assessment, business storytelling, recommendation design and responsible AI. [cite]turn0search0[

The central business scenario concerns Geldium Finance, a financial-services provider experiencing credit-card delinquency. The project objective is to use data-driven and AI-assisted methods to understand delinquency risk, design predictive logic and support proactive intervention strategies.

The supplied work material shows progression through the introductory scenario, exploratory data analysis and risk profiling, and the predictive-modeling phase. Major learning areas include dataset understanding, missing-data treatment, risk-factor identification, synthetic data considerations, predictive model selection, evaluation, explainability and fairness.

Because the supplied evidence consists primarily of the simulation instructions and learning screens rather than the underlying Geldium dataset or a final model-output file, this report distinguishes project requirements and methodology from numerical model results. No unsupported accuracy figures or dataset statistics are invented.

1. PROGRAM / ORGANIZATION OVERVIEW

Tata iQ is presented in the simulation as Tata Group's advanced analytics and AI center of excellence. The Forage program places the learner in a simulated financial-services analytics team and uses a fictional client, Geldium Finance, to provide a realistic business context.

The official Forage program page describes the experience as a self-paced job simulation designed to replicate work performed by an AI transformation consultant. It lists skills including exploratory data analysis, data quality management, predictive analytics, model selection, model validation, ethical reasoning, business communication and strategic thinking.

[cite]turn0search0[

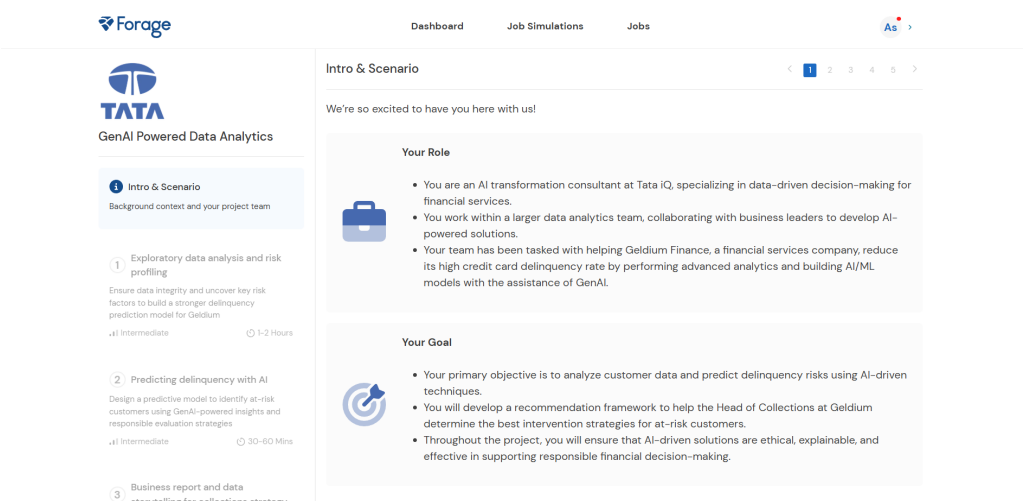


Figure 1. Introductory scenario and role description from the supplied evidence.

2. BUSINESS SCENARIO AND OBJECTIVES

The simulated assignment begins with a rise in credit-card delinquency at Geldium Finance, with customers missing payments beyond the 30-day threshold. The existing process relies on historical trends and manual case handling. The proposed analytics approach is intended to help identify at-risk customers and support more targeted interventions.

The main objectives documented in the simulation are:

- Review customer data to identify patterns, data-quality issues and risk factors.
- Use GenAI-assisted analytics to support exploratory analysis and predictive-model development.
- Develop predictive logic for identifying customers at higher risk of delinquency.
- Provide structured recommendations for intervention strategies.
- Ensure AI-driven risk assessment remains ethical, explainable and responsible.

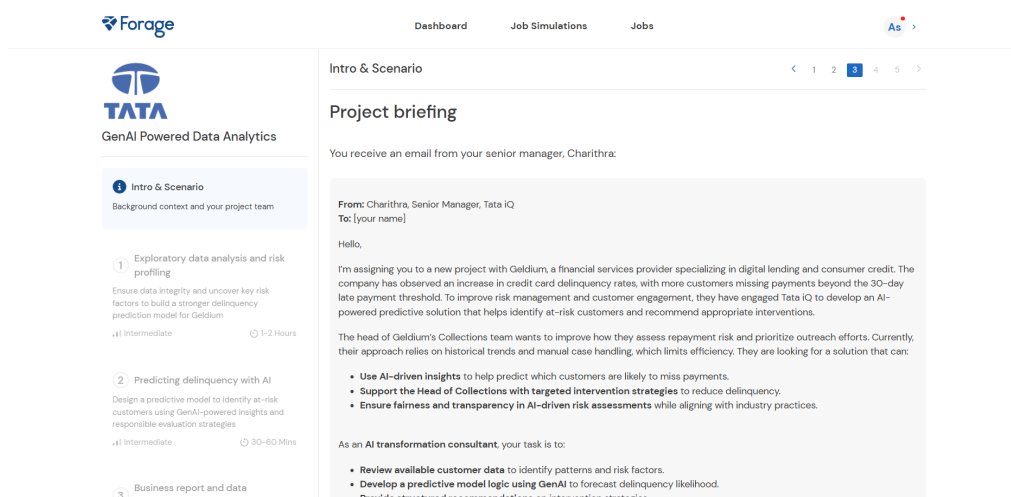


Figure 2. Project briefing supplied in the simulation.

3. ROLE AND RESPONSIBILITIES

The learner is assigned the simulated role of AI Transformation Consultant within Tata iQ's Financial Services Data & AI Strategy team. The supplied screen identifies Ayush Shekhar in this role and describes responsibilities around customer-data analysis, predictive modeling and AI-driven recommendations.

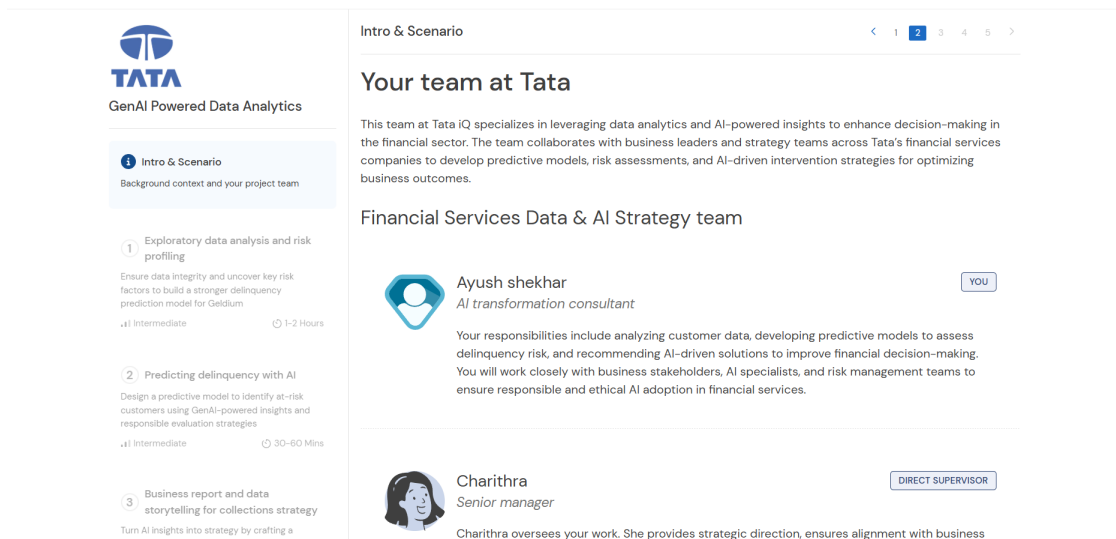


Figure 3. Simulated team and role assignment.

Key responsibilities

- Analyze customer data and understand variables relevant to delinquency.
- Develop predictive-model logic and consider appropriate modeling techniques.
- Communicate findings in a business-oriented form.
- Work with the concepts of fairness, transparency and explainability.

4. METHODOLOGY / WORKFLOW

The internship simulation follows a progressive analytics lifecycle:

Phase	Main activity	Output / purpose
1	Understand scenario	Define business problem and objectives
2	EDA & risk profiling	Assess structure, completeness, anomalies and risk indicators
3	Data quality treatment	Select appropriate missing-data handling strategies
4	Risk-factor analysis	Explore relationships between variables and delinquency
5	Predictive modeling	Select model approach and define evaluation plan
6	Responsible AI	Address fairness, explainability, validation and human oversight
7	Business communication	Translate analytics into recommendations and reporting

This workflow matches the simulation's transition from exploratory data analysis to predictive modeling and business-focused decision support. [cite]turn0search0[

5. TASK 1 — EXPLORATORY DATA ANALYSIS AND RISK PROFILING

Task 1 focuses on establishing a reliable analytical foundation before predictive modeling. The learner is instructed to understand the dataset structure, identify missing or inconsistent data, detect anomalies and recognize early indicators that could be relevant to delinquency prediction.

The task specifically introduces:

- Dataset familiarization: key variables, categorical versus numerical fields, and data completeness.
- Missing-value and outlier identification.
- Relationships between variables, such as credit utilization, income and delinquency.
- Detection of patterns and risk factors.
- Use of GenAI for summarization and exploratory assistance rather than blindly replacing analytical judgment.

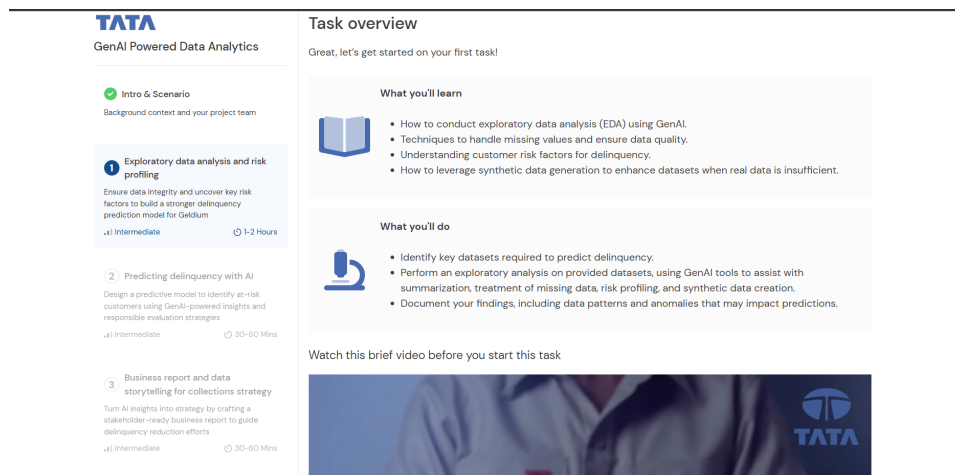


Figure 4. Task 1 overview from the supplied simulation screen.

6. DATA QUALITY AND MISSING-VALUE HANDLING

A major theme of the simulation is that predictive models should not be built on unreliable data. The material explains that missing values may arise from random system errors, skewed data collection or customer behavior. It emphasizes understanding the reason for missingness before selecting a treatment.

Approaches covered in the simulation include:

- Deletion: remove incomplete observations or columns when missingness is limited or a field cannot be meaningfully used.
- Imputation: use mean, median, mode, forward/backward filling or predictive approaches where appropriate.
- Missingness analysis: consider MCAR, MAR and MNAR patterns before filling values.
- GenAI assistance: use AI to detect gaps and suggest strategies, while validating the final decision using statistical and domain reasoning.

The screenshot displays the TATA GenAI Powered Data Analytics interface. On the left, a sidebar lists four steps: 1. Intro & Scenario (1-2 Hours), 2. Exploratory data analysis and risk profiling (30-60 Mins), 3. Predicting delinquency with AI (30-60 Mins), and 4. Business report and data storytelling for collections strategy (30-60 Mins). The main content area is titled 'Techniques to handle missing values and ensure data quality'. It explains that missing values can skew predictions and lead to incorrect risk assessments. It then lists common causes of missing data: Random errors, Skewed data collection, and Customer behavior. Finally, it provides three methods to handle missing values: Deleting missing data, Imputation (replacing missing values), and AI-Assisted Imputation.

TATA
GenAI Powered Data Analytics

1. Intro & Scenario
Background context and your project team
1-2 Hours

2. Exploratory data analysis and risk profiling
Ensure data integrity and uncover key risk factors to build a stronger delinquency prediction model for Geldum
30-60 Mins

3. Predicting delinquency with AI
Design a predictive model to identify at-risk customers using GenAI-powered insights and responsible evaluation strategies
30-60 Mins

4. Business report and data storytelling for collections strategy
Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts
30-60 Mins

Techniques to handle missing values and ensure data quality

Now that you understand how EDA helps assess dataset completeness, the next step is addressing missing values to ensure reliable insights. In financial risk analytics, incomplete or inconsistent data can skew predictions and lead to incorrect risk assessments. When handling missing data, it is critical to first analyze **the reason for missingness**—whether it is random, systematic, or indicative of underlying biases. While GenAI models can assist in detecting patterns and suggesting imputation strategies, statistical techniques (e.g., mean, median, or regression-based imputation) remain the industry standard due to their transparency and reproducibility. Blindly applying GenAI-generated imputation without understanding data context can introduce significant bias.

What are the common causes of missing data?

Missing data can occur due to:

- **Random errors** (e.g., a system glitch fails to record a payment).
- **Skewed data collection** (e.g., high-income customers are less likely to disclose salary details).
- **Customer behavior** (e.g., financially distressed individuals may avoid reporting debt).

Understanding the cause helps determine the best approach for handling missing values.

How to handle missing values:

1. **Deleting missing data** – If only a **small percentage** of data is missing, removing incomplete entries may be the best approach. However, this can reduce the sample size.
2. **Imputation (replacing missing values)** –
 - **Mean, median, or mode imputation** fills gaps with typical values.
 - **Forward or backward filling** uses existing data trends to estimate missing entries.
3. **AI-Assisted Imputation** –

Figure 5. Simulation guidance on causes and handling of missing data.

7. CUSTOMER RISK FACTORS AND SYNTHETIC DATA

The simulation identifies several customer-level factors that can be examined when assessing delinquency risk. These are presented as analytical indicators rather than as a final causal conclusion.

- **Payment history:** previous late or missed payments are an important historical indicator.
- **Credit utilization:** high usage of available credit can indicate financial pressure.
- **Debt-to-income ratio:** a higher ratio may indicate greater difficulty managing obligations.
- **Recent credit activity:** sudden increases in new credit accounts or applications may require additional analysis.
- **Employment and income stability:** changes or inconsistency may be relevant to repayment risk.

The simulation also introduces synthetic data generation as a way to supplement limited or incomplete datasets. It stresses validation against real-world distributions and the use of synthetic data as a supplementary rather than unquestioned primary source.


GenAI Powered Data Analytics

✓ Intro & Scenario

Background context and your project team

1 Exploratory data analysis and risk profiling

Ensure data integrity and uncover key risk factors to build a stronger delinquency prediction model for Geldium

✓(1) Intermediate 1-2 Hours

2 Predicting delinquency with AI

Design a predictive model to identify at-risk customers using GenAI-powered insights and responsible evaluation strategies

✓(1) Intermediate 30-60 Mins

3 Business report and data storytelling for collections strategy

Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts

✓(1) Intermediate 30-60 Mins

How to leverage synthetic data generation to enhance datasets

In financial services, incomplete or inconsistent data can make it difficult to build reliable predictive models. If key information is missing—such as payment history or income details—delinquency risk assessments may become inaccurate. When real-world data is limited, sensitive, or incomplete, **synthetic data generation** can help fill gaps, simulate scenarios, and improve dataset quality while maintaining privacy and compliance.

Synthetic data is **artificially generated data** that mimics real-world data patterns. Instead of using actual customer records, synthetic data is created using [statistical models](#) or [AI-driven techniques](#) to supplement missing values or expand datasets for testing. This ensures that no real customer information is exposed while still preserving the integrity of the analysis.

While synthetic data generation can be useful for filling in gaps and expanding datasets, it should be applied with caution in financial risk modeling. Traditional statistical simulation techniques such as [Monte Carlo simulations](#), [bootstrapping](#), and [probabilistic modeling](#) are often preferred due to their explainability, reproducibility, and ability to align with industry regulations.

GenAI-generated synthetic data should be:

- ✓ **Validated against real-world distributions** to ensure accuracy.
- ✓ **Cross-checked with statistical models** to prevent introducing artificial patterns or biases.
- ✓ **Used as a supplementary tool**, not a primary data source, especially in regulatory environments.

When to use synthetic data in financial services:

- **Enhancing small datasets** – If real customer data is limited, synthetic records can supplement training data for AI models.
- **Filling in missing data** – When real data is incomplete, synthetic data can approximate realistic values based on existing patterns.
- **Testing AI models** – Before deploying AI risk models, synthetic data can be used to simulate different delinquency scenarios.

Figure 6. Guidance on synthetic data generation.

8. TASK 1 — EXPECTED ANALYTICAL OUTPUT

The task instructions require the learner to document findings in a concise report. Based on the supplied screens, the expected reporting structure is:

- Notable missing or inconsistent data.
- Key anomalies or unusual patterns.
- Early indicators of delinquency risk.
- A short narrative summarizing initial data-quality observations.
- A table of 2–3 missing-data issues, chosen treatment methods and one-line justifications.
- A list of high-risk indicators with explanations of why they matter.

The screenshot displays a simulation interface with a sidebar on the left and a main content area on the right. The sidebar contains a list of tasks, with the fourth task, 'Implementing an AI-driven collections strategy', highlighted. The main content area is titled 'Prompts to try:' and contains several sections of text and bullet points. The first section lists two prompts: 'Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modelling delinquency.' and 'Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.' The second section, titled 'Action:', contains a paragraph: 'Document your findings in bullet points for your report. Focus on:' followed by three bullet points: 'Notable missing or inconsistent data', 'Key anomalies', and 'Early indicators of delinquency risk'. The third section, titled 'Step 2: Address missing data and data quality issues', contains a paragraph: 'Now that you've identified gaps, it's time to decide how to handle missing data to maintain accuracy.' followed by a section titled 'What to do:' which lists three bullet points: 'Identify gaps or missing values in critical features (e.g., payment history, income, credit utilization).', 'Determine the best treatment approach for each case:', and a sub-section with three bullet points: 'Remove: Drop columns with excessive missing data.', 'Impute: Fill in missing values using mean, median, or predictive modeling.', and 'Generate synthetic data: Use AI tools to create realistic values while maintaining fairness and distribution patterns.' The final bullet point is 'Use GenAI to assist with suggesting strategies as well as automating parts of the imputation process'.

Intermediate 30-60 Mins

4 Implementing an AI-driven collections strategy

Design an AI-powered collections strategy to help Celdum engage high-risk customers with smart, fair, and scalable outreach

Intermediate 30-60 Mins

Prompts to try:

- Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modelling delinquency.
- Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.

Action:

Document your findings in bullet points for your report. Focus on:

- Notable missing or inconsistent data
- Key anomalies
- Early indicators of delinquency risk

Then, write a short paragraph (3–5 sentences) summarizing your initial data quality observations.

Step 2: Address missing data and data quality issues

Now that you've identified gaps, it's time to **decide how to handle missing data** to maintain accuracy.

What to do:

- Identify gaps or missing values in critical features (e.g., payment history, income, credit utilization).
- Determine the best treatment approach for each case:
 - **Remove:** Drop columns with excessive missing data.
 - **Impute:** Fill in missing values using **mean, median, or predictive modeling**.
 - **Generate synthetic data:** Use AI tools to create realistic values while maintaining fairness and distribution patterns.
- Use GenAI to assist with suggesting strategies as well as automating parts of the imputation process

Prompts to try:

Figure 7. Example prompts and reporting instructions supplied by the simulation.

9. TASK 2 — PREDICTING DELINQUENCY WITH AI

Task 2 moves from exploration to prediction. The objective is to use GenAI-assisted reasoning to design a predictive approach that can identify customers at risk of missing payments and support proactive outreach.

The simulation introduces common predictive approaches such as decision trees, logistic regression and neural networks. It also emphasizes that GenAI can help draft model logic or code and interpret results, but generated outputs must be validated and refined by analysts.

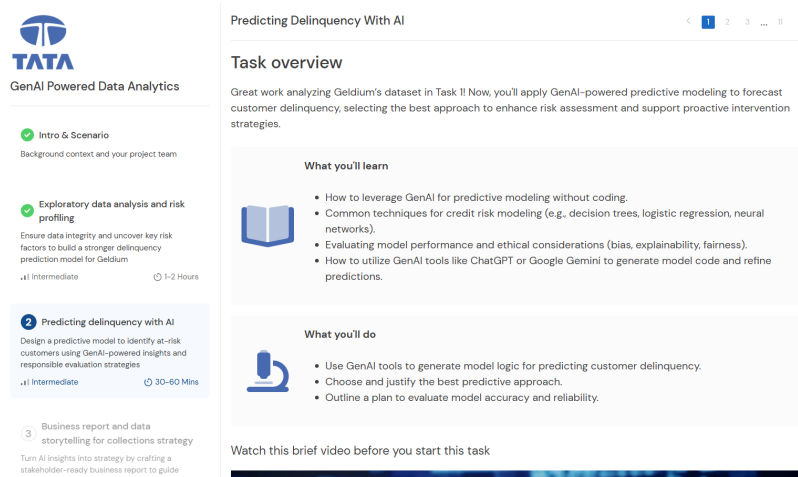


Figure 8. Task 2 overview and learning objectives.

Predictive-model workflow

- Define the target outcome: delinquency / missed payment.
- Prepare and validate features identified during EDA.
- Select a suitable model based on data characteristics and business requirements.
- Evaluate accuracy and reliability using appropriate validation methods.
- Check fairness, explainability and operational suitability before use.

10. MODEL SELECTION, VALIDATION AND RESPONSIBLE AI

The supplied material does not contain a completed model, numerical performance metrics or a final model artifact. Therefore, this report does not claim a specific trained-model accuracy. Instead, it records the evaluation framework taught by the simulation.

Model-selection considerations

- Logistic regression: useful as an interpretable baseline for binary delinquency prediction.
- Decision trees: can provide intuitive rule-like decision paths and capture nonlinear relationships.
- Neural networks: can model complex patterns but may require additional care for explainability and validation.

Evaluation considerations

- Use a hold-out or cross-validation strategy appropriate to the dataset.
- Assess relevant classification metrics rather than relying on accuracy alone.
- Check calibration and error patterns where risk scores will drive interventions.
- Review subgroup performance and potential fairness concerns.
- Use explainability methods and human review for consequential decisions.

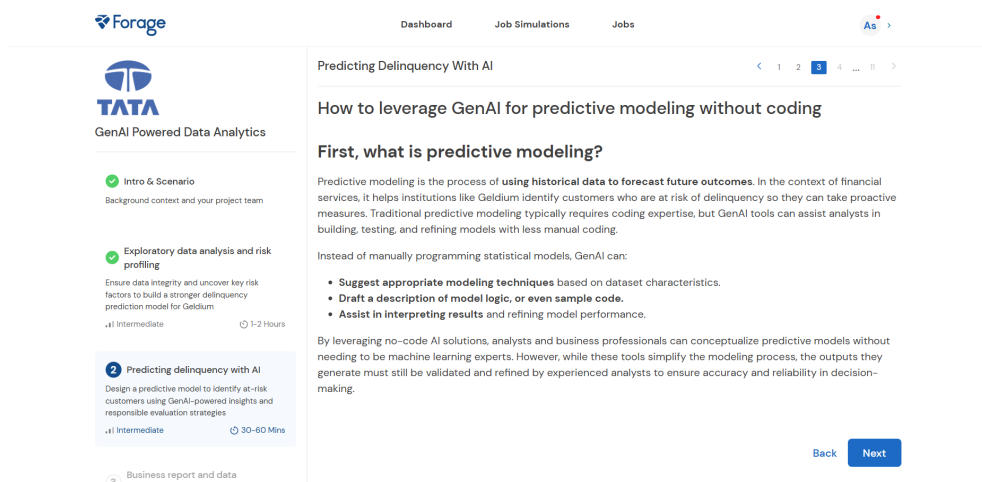


Figure 9. GenAI-assisted predictive-modeling guidance.

11. USE OF GENERATIVE AI AS AN ANALYTICAL ASSISTANT

A defining feature of the simulation is the use of GenAI to accelerate analytical work. The material presents GenAI as an assistant for summarization, feature exploration, prompt-driven analysis, model-logic drafting and interpretation—not as a substitute for validation.

Examples of GenAI-supported activities covered in the supplied material:

- Summarize key columns, patterns, outliers and missing values.
- Suggest questions for EDA and potential risk indicators.
- Recommend possible imputation strategies.
- Explain relationships between variables in natural language.
- Generate or refine predictive-model logic and sample code.
- Assist with communicating analytical insights to stakeholders.

Responsible use principles

- Do not expose sensitive financial information unnecessarily.
- Validate AI-generated recommendations with statistics and domain expertise.
- Do not treat synthetic data as automatically representative of real customers.
- Maintain fairness, transparency and explainability.
- Keep human oversight in decisions that affect customers.

12. SKILLS AND LEARNING OUTCOMES

The official Forage listing identifies skills including AI analytics, data interpretation, data quality management, exploratory data analysis, model selection, model validation, predictive analytics, ethical reasoning and strategic thinking. [cite]turn0search0[

Technical / Analytical	Business / Professional
Exploratory Data Analysis	Problem framing
Data quality assessment	Business communication
Missing-data strategies	Stakeholder-oriented reporting
Predictive modeling concepts	Strategic thinking
Model selection & validation	Responsible decision-making
GenAI-assisted analytics	Ethical reasoning

Overall learning outcome: the simulation provided an end-to-end view of how a data/AI project can move from problem definition and data quality through modeling, evaluation and business communication.

13. CONCLUSION AND FUTURE SCOPE

The Tata iQ GenAI Powered Data Analytics simulation provided a practical introduction to applying data analytics and AI concepts to a financial-services problem. The experience connected exploratory analysis, data quality, risk profiling, predictive modeling and responsible AI within one continuous business scenario.

A key learning from the project is that predictive modeling is not only a model-building exercise. Reliable results depend on understanding the dataset, handling missing and inconsistent values carefully, identifying meaningful risk indicators, validating model behavior and communicating results clearly to decision-makers.

For future work, the project can be extended by obtaining the actual permitted sample dataset, implementing the predictive model in Python, performing quantitative EDA, comparing multiple classifiers, applying robust validation and calibration, documenting subgroup fairness metrics, and developing a dashboard for risk monitoring and intervention tracking.

14. REFERENCES

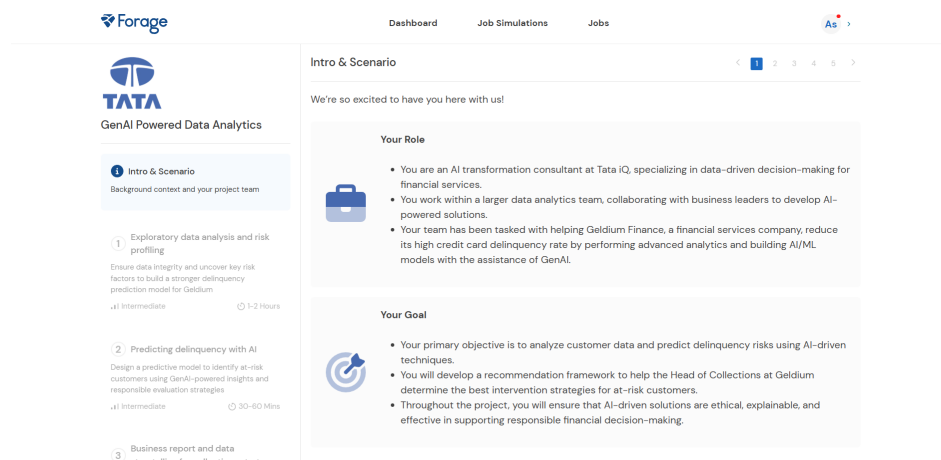
1. Forage — Tata GenAI Powered Data Analytics Job Simulation. Official program page. `[cite\turn0search0]`
2. Tata Insights and Quants / Forage — program announcement describing the simulation as a practical experience covering predictive modeling, data exploration, business storytelling and responsible AI. `[cite\turn0search3]`
3. User-provided screenshots of the Forage simulation, used as the primary evidence for task descriptions and learning material in this report.

Academic note: This report intentionally avoids claiming numerical dataset findings or model-performance values that were not present in the supplied evidence.

APPENDIX A — ACTIVITY EVIDENCE

Selected screenshots supplied as evidence of the learning path and project activities.

A1. Program overview



A2. EDA task overview



GenAI Powered Data Analytics

Task overview

Great, let's get started on your first task!

1 Intro & Scenario

Background context and your project team

2 Exploratory data analysis and risk profiling

Ensure data integrity and uncover key risk factors to build a stronger delinquency prediction model for Geddium

⌕ Intermediate ⌚ 1-2 Hours

3 Predicting delinquency with AI

Design a predictive model to identify at-risk customers using GenAI-powered insights and responsible evaluation strategies

⌕ Intermediate ⌚ 30-60 Mins

4 Business report and data storytelling for collections strategy

Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts

⌕ Intermediate ⌚ 30-60 Mins

What you'll learn

- How to conduct exploratory data analysis (EDA) using GenAI.
- Techniques to handle missing values and ensure data quality.
- Understanding customer risk factors for delinquency.
- How to leverage synthetic data generation to enhance datasets when real data is insufficient.

What you'll do

- Identify key datasets required to predict delinquency.
- Perform an exploratory analysis on provided datasets, using GenAI tools to assist with summarization, treatment of missing data, risk profiling, and synthetic data creation.
- Document your findings, including data patterns and anomalies that may impact predictions.

Watch this brief video before you start this task



A3. Missing-data guidance



GenAI Powered Data Analytics

Intro & Scenario

Background context and your project team

1

Exploratory data analysis and risk profiling

Ensure data integrity and uncover key risk factors to build a stronger delinquency prediction model for Goldium

Intermediate

1-2 Hours

2

Predicting delinquency with AI

Design a predictive model to identify at-risk customers using GenAI-powered insights and recommendation strategies

Intermediate

30-60 Mins

3

Business report and data storytelling for collections strategy

Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts

Intermediate

30-60 Mins

Techniques to handle missing values and ensure data quality

Now that you understand how EDA helps assess dataset completeness, the next step is addressing missing values to ensure reliable insights. In financial risk analysis, incomplete or inconsistent data can skew predictions and lead to incorrect risk assessments. When handling missing data, it is critical to first analyze the **reason for missingness**—whether it is random, systematic, or indicative of underlying biases. While GenAI models can assist in detecting patterns and suggesting imputation strategies, statistical techniques (e.g., mean, median, or regression-based imputation) remain the industry standard due to their transparency and reproducibility. Blindly applying GenAI-generated imputation without understanding data context can introduce significant bias.

What are the common causes of missing data?

Missing data can occur due to:

- **Random errors** (e.g., a system glitch fails to record a payment).
- **Skewed data collection** (e.g., high-income customers are less likely to disclose salary details).
- **Customer behavior** (e.g., financially distressed individuals may avoid reporting debt).

Understanding the cause helps determine the best approach for handling missing values.

How to handle missing values:

1. **Deleting missing data** – If only a **small percentage** of data is missing, removing incomplete entries may be the best approach. However, this can reduce the sample size.

2. Imputation (replacing missing values) –

- **Mean, median, or mode imputation** fills gaps with typical values.
- **Forward or backward filling** uses existing data trends to estimate missing entries.

- ### 3. AI-Assisted Imputation –

A4. Risk-factor guidance



GenAI Powered Data Analytics

 **Intro & Scenario**

Background context and your project team



Exploratory data analysis and risk profiling

Ensure data integrity and uncover key risk factors to help storage delinquency prediction model for Geldium

 Intermediate

 1-2 Hours



Predicting delinquency with AI

Design a predictive model to identify at-risk customers using GenAI-powered Insights and analytics evaluation strategies

 Intermediate

 30-60 Mins



Business report and data storytelling for collections strategy

Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts

 Intermediate

 30-60 Mins

Understanding customer risk factors for delinquency

Now that you've learned how to ensure data quality and handle missing values, the next step is identifying key customer risk factors that influence credit card delinquency. Understanding these factors will help refine predictive models and improve intervention strategies.

Why risk factors matter in financial decision-making:

Lenders assess various financial and behavioral indicators to determine whether a customer is likely to miss payments. By analyzing these factors, Geldium's Collections team can proactively identify customers who may need early intervention, reducing financial losses and improving repayment outcomes.

Key risk factors for delinquency

- **Payment history** – Customers with a history of late or missed payments are more likely to default.
- **Credit utilization rate** – High usage of available credit can indicate financial stress and potential repayment issues.
- **Debt-to-income (DTI) ratio** – A high DTI suggests a customer may struggle to manage their financial obligations.
- **Recent credit activity** – A sudden increase in new credit accounts or loan applications may signal financial instability.
- **Employment and income stability** – Frequent job changes or inconsistent income can contribute to a higher risk of missed payments.
- **Demographic trends** – While AI models must avoid bias, certain patterns (e.g., younger customers with limited credit history) may require additional analysis.

AI-driven models analyze multiple risk factors simultaneously, detecting patterns that may not be immediately obvious. However, ensuring that these models remain **fair and explainable** is crucial to avoiding biases in decision-making.

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APPENDIX B — PREDICTIVE MODELING EVIDENCE

These screens show the transition from EDA to predictive modeling and the role of GenAI in the simulation.



GenAI Powered Data Analytics

1

Intro & Scenario

Background context and your project team

2

Exploratory data analysis and risk profiling

Ensure data integrity and uncover key risk factors to build a stronger delinquency prediction model for Geldium

Intermediate 1-2 Hours

3

Predicting delinquency with AI

Design a predictive model to identify at-risk customers using GenAI-powered insights and responsible evaluation strategies

Intermediate 30-60 Mins

4

Business report and data storytelling for collections strategy

Turn AI insights into strategy by crafting a stakeholder-ready business report to guide delinquency reduction efforts

Intermediate 30-60 Mins

Predicting Delinquency With AI

Task overview

Great work analyzing Geldium's dataset in Task 1! Now, you'll apply GenAI-powered predictive modeling to forecast customer delinquency, selecting the best approach to enhance risk assessment and support proactive intervention strategies.

What you'll learn

- How to leverage GenAI for predictive modeling without coding.
- Common techniques for credit risk modeling (e.g., decision trees, logistic regression, neural networks).
- Evaluating model performance and ethical considerations (bias, explainability, fairness).
- How to utilize GenAI tools like ChatGPT or Google Gemini to generate model code and refine predictions.

What you'll do

- Use GenAI tools to generate model logic for predicting customer delinquency.
- Choose and justify the best predictive approach.
- Outline a plan to evaluate model accuracy and reliability.

Watch this brief video before you start this task



Let's get started

You're back in a virtual check-in with Charithra to discuss the next phase of your work with Geldium. After completing the report, you now have a clearer understanding of the dataset—its gaps, inconsistencies, and key risk indicators. Now, it's time to move beyond exploration and into prediction.

"Geldium wants to use AI to predict which customers are at risk of missing payments so they can take proactive steps to reduce delinquency. Our goal is to develop a model that supports the Collections team in prioritizing outreach and intervention strategies."

Your role in this phase is to:

- Use **GenAI tools** to develop a **predictive model** for identifying high-risk customers.
- Choose and **justify the best approach**—whether decision trees, logistic regression, or another technique.
- Define a plan to **evaluate model performance** while ensuring fairness and explainability.

Geldium will rely on this model to enhance risk assessment and guide intervention decisions. Your structured approach will help determine how accurate and reliable the AI-driven predictions will be.

"Think critically about your approach," Charithra adds. "Explain why you selected a particular method and how we'll ensure the model performs well in real-world conditions. This isn't just about prediction—it's about making responsible, data-driven decisions."

It's time to take the insights from your EDA findings and transform them into an actionable AI-driven delinquency prediction model.

[Back](#) [Next](#)

Figure B1-B2. Task 2 and predictive-modeling guidance.